Imad Serhal

Control Systems Engineer | Model-Based Control | Electric Powertrains

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PROFESSIONAL SUMMARY

Control Systems Engineer with an M.Sc. in Automation & Control Systems Engineering and hands-on experience in model-based control, electric drives, CAN diagnostics, prototype validation, and electromechanical systems.

Advanced MATLAB/Simulink background developed through electric-powertrain research and an M.Sc. thesis focused on control of an Electrically Excited Synchronous Motor for a heavy-duty IVECO vehicle. Experience developing linear and nonlinear dynamic models, designing and tuning cascaded control loops, implementing FOC, MTPA, flux weakening and MTPV strategies, and validating controller robustness under changing load conditions.

Currently working on autonomous electric platforms, analysing closed-loop behaviour, CAN signals, torque response, braking behaviour, system limits and controller performance on physical prototype machines.

TECHNICAL SKILLS

Control Systems: Model-Based Control, PID/PI Control, Cascaded Control Loops, Anti-Windup, State-Space Control, Control System Tuning

MATLAB / Simulink: Dynamic Modelling, Nonlinear Simulation, Control Algorithm Development, Lookup Tables, Sensitivity Analysis, Numerical Optimisation

Electric Powertrain Control: Field-Oriented Control (FOC), MTPA, MTPV, Flux Weakening, Torque Control, Voltage/Current Constraints, Electric Drives

Modelling: d-q Motor Models, Flux Maps, Magnetic Saturation, Cross-Coupling, Mechanical Dynamics, System Identification

Vehicle / System Integration: CAN J1939, CANopen, CAN Diagnostics, Sensor/Actuator Integration, Data Logging, Prototype Testing, Fault Analysis

Embedded & Software: C/C++, Python, Real-Time Control Systems, Embedded Systems

State Estimation: Kalman Filtering, Sensor Fusion, Dynamic-System Estimation

Testing & Validation: Closed-Loop Response Analysis, Electric-Drive Testing, Performance Validation, Fault Handling, System Robustness

PROFESSIONAL EXPERIENCE

Control Systems Engineer, Robotics Testing & Validation Apr 2025 - Present
STIGA Italy

- Analyse and validate closed-loop motor and vehicle-motion behaviour on autonomous electric platforms, including speed tracking, torque response, braking behaviour and stability under changing load conditions.
- Perform real-time CAN-bus diagnostics and signal analysis to evaluate controller performance, investigate abnormal behaviour and support firmware and control-strategy optimisation.
- Work directly with electric-drive hardware including traction motors, motor controllers, batteries, sensors, actuators and embedded computing systems during prototype development and validation.
- Execute system-level test campaigns to evaluate control limits, fault handling, transient response and robustness under representative operating conditions.
- Analyse experimental data using engineering scripts and data-processing tools to correlate commanded behaviour with measured system response and support root-cause investigation.
- Support integration and debugging of motors, controllers, LiDAR sensors and NVIDIA Jetson computing units on autonomous prototype platforms.

Research Trainee, Advanced Motor Control Feb 2024 - Jun 2024

University of Padova Italy

- Developed linear and nonlinear control-oriented models of IPM, SPM and EESM electric machines, accounting for excitation dynamics, magnetic saturation and cross-coupling effects.
- Designed and tuned cascaded current and speed control loops with PI control, anti-windup and bandwidth separation.
- Developed control-oriented flux, torque and speed maps to evaluate operating envelopes and controller performance.
- Investigated electric-drive behaviour across varying speeds, torque demands and electrical constraints.

Mechanical Design Engineer Jun 2021 - Sep 2022

My Projects SAL Lebanon

- Performed engineering calculations and mechanical system design for HVAC and pumping systems, strengthening practical understanding of physical systems, fluid behaviour and mechanical integration.
- Produced AutoCAD technical drawings and engineering documentation and supported installation activities on site.

PROJECTS

M.Sc. Thesis: Control of an Electrically Excited Synchronous Motor for a Heavy-Duty IVECO Vehicle

- Developed complete linear and nonlinear EESM models in MATLAB/Simulink using d-q equations, flux-linkage maps and saturation/cross-coupling characteristics.
- Designed and tuned cascaded speed, stator-current and excitation-current controllers with anti-windup and voltage/current constraint handling.
- Implemented Field-Oriented Control with MTPA, flux weakening and MTPV strategies to cover the complete torque-speed operating envelope.
- Developed logic for smooth transitions between MTPA, flux-weakening and MTPV operating regions according to speed, torque and voltage constraints.
- Evaluated controller performance through transient simulations at multiple speeds and torque loads, including high-load operating conditions.
- Developed lookup-table based control strategies to reduce real-time computational requirements while maintaining optimal current references.
- Applied Extremum Seeking Control for online optimisation of excitation current, achieving up to 10% reduction in simulated Joule losses.
- Validated stability and robustness by varying load conditions and introducing modelling errors into the system.

Autonomous GPS-Guided Robotic Platform

- Built GPS waypoint navigation on an Arduino Mega using bearing and Haversine distance calculations.
- Integrated and calibrated an MPU9255 magnetometer for heading feedback, steering toward each waypoint with a heading-error deadband; added Bluetooth commands and ultrasonic obstacle sensing.

EDUCATION

M.Sc. Automation & Control Systems Engineering 2022 - 2024

University of Padova, Italy

Relevant coursework: Advanced Control, Nonlinear Control, Digital Control, Robotics, Embedded Systems, System Identification, Machine Learning

B.Eng. Mechanical Engineering 2017 - 2022

Beirut Arab University, Lebanon

Relevant coursework: Control Theory, System Dynamics, Thermodynamics, Fluid Mechanics

LANGUAGES

English: C1 (Fluent) | Arabic: Native | Italian: A2 (Improving)